Converting Mass, Area and Volume

Question Bank - Additional 5.08 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, convert with the ratio given.

1. $1\text{ lb} = 0.454\text{ kg}$. 40 lb to kilograms.

18.16 kg

(40 ÷ 2.204 = 18.15 by the other ratio.)

2. $1\text{ kg} = 2.204\text{ lb}$. 5 kg to pounds.

11.02 lb

3. $1\text{ gal} = 3.785\text{ L}$. 12 gal to liters.

45.42 L

4. $1\text{ L} = 0.264\text{ gal}$. 50 L to gallons.

13.2 gal

For questions 5 through 7, say how many times the ratio is used.

5. $1\text{ yd} = 3\text{ ft}$. 7 yd^2 to square feet.

63 ft²

7 × 9: twice.

6. $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$. 40 fl oz to liters.

1.2 L

5 c × 0.24.

7. $1\text{ ft} = 12\text{ in}$. 4 ft^3 to cubic inches.

6,912 in³

4 × 1,728: three times.

Choose one.

8. $1\,\text{m} = 100\,\text{cm}$. A floor is $3\,\text{m}^2$ in area. How many square centimeters is that?

- ○ A. $600\,\text{cm}^2$
- ○ B. $10{,}000\,\text{cm}^2$
- ○ C. $300\,\text{cm}^2$
- ● **D.** $30{,}000\,\text{cm}^2$

D

3 × 10,000.